

DATA SCIENCE

DURATION- 6 Month

1. DATA SCIENCE AND VISUALIZATION WITH PYTHON

PYTHON BASICS

- What is Python
- Application of Python
- Why use Python for AI-ML Installation Anaconda/Other Idle
- Python Tokens
- Data Types in Python
- Conditional Statement
- Loops in Python
- Functions in Python
- Advance Functions
- File Handling

Module-2: Python NumPy & Pandas

- Introduction to NumPy
- Exploring a NumPy Array
- Indexing & Slicing a NumPy Array
- Manipulating a NumPy Array
- Performing Mathematical & Statistical Functions using NumPy
- Performing Linear Algebra Operations using NumPy
- Introduction to Pandas
- Exploring Pandas Series
- Introduction to Pandas DataFrame
- Importing & Exporting Data
- Implementing basic DataFrame functionalities
- Exploring Descriptive Statistics with Pandas

Modules -3: EDA

- Data Types
- Dispersion & Skewness

- Data imputation
- Data Pre-processing
- Data Cleaning
- Data Manipulation
- Advanced Manipulation

Module 4- SQL for Data Analytics

- SQL Basic
- SQL Joins
- SQL Aggregations
- Subqueries and Temp Tables
- SQL Data Cleaning
- Window Functions

Module-5: Data Visualization

- Why Data Visualization?
- Introduction to Data Visualization
- Libraries & Tools for Data Visualization in Python
- Static Data Visualization Using Seaborn
- Interactive Data Visualization Using Plotly Express
- Interactive Animations & Facet Plots

Mini Test

Mini Assignment

Module 6: Tableau

- Understanding Data
- Creating Your First visualization
- Tableau Calculations
- Formatting Visualizations
- Manipulating Data in Tableau
- Creating Dashboards AND Stories
- Distributing & Publishing Your Visualization

Module 08 - Visual Storytelling using Power BI

- Introduction To Power BI
- Creating Power BI Reports, Auto Filters
- Report Visualization And Properties
- Chart And Map Report Properties
- Hierarchies And Drilldown Reports

- Power BI Deployment & Cloud
- Improving Power BI Reports and More...

2. MATHS AND APPLIED STATISTICS

Module-1: Mathematics for Machine Learning

- Linear Algebra
- Introduction to Calculus
- Multivariable Calculus

Module-2: Statistical Foundations

- Applications of Statistics
- Introduction to Statistics
- Categories of Data
- Basic Terminologies in Statistics
- Sampling techniques
- Descriptive Statistics
- Measure Used in Descriptive Statistics
- Z-Scores

Module-3: Probability

- What is Probability?
- Rules of probability
- Types of Probability
- Random Variables
- Probability Distribution Functions

Module-4: Inferential Statistics

- Introduction to Inferential Statistics
- Hypothesis Testing
- Normal Distribution
- P-value
- One-tailed and Two-tailed tests
- One Sample Z test
- One Sample T test
- Independent Sample T test
- Chi-square test
- ANOVA

3. MACHINE LEARNING

Module-1: Introduction to Machine Learning

- What is Machine Learning?
- Applications of Machine Learning
- Machine Learning in your daily life
- Machine Learning in Retail
- Steps Involved in Machine Learning

Module-2: Regression

- Introduction to Regression
- Linear Regression
- Evaluation Metrics in Regression Models
- Logistic Regression

Module-3: Supervised Classification

- Why Use Classification?
- Application of Classification Algorithms
- Introduction to Classification
- Types of Classification Algorithms
- Classification: Decision Tree
- Classification: Random Forest
- ML in Banking & Finance - Benefits
- Classification: SVM
- Classification: KNN
- Classification: Naïve Bayes
- Evaluating Classification Models
- Model Optimization Techniques
- Model Boosting Techniques
- Introduction to PyCaret
- Dealing with Unbalanced Datasets

Module-4: Association Rules Mining

- What are Association Rules?
- Association Rule Parameters
- A-priori Algorithm
- Market Basket Analysis

Module-5: Unsupervised Learning

- What is Unsupervised Learning?

- Application of Unsupervised Learning
- Introduction to Clustering
- Types of Clustering
- Partitioning Methods: K-means, DBSCAN, Spectral
- Hierarchical Methods: Hierarchical

Module- 6: Dimension Reduction

- PCA
- Factor Analysis
- LDA

Module-7: Recommendation System

- What is a Recommendation System?
- Need for a Recommendation System
- Recommendation System Use Cases
- Applications of Recommendation System
- Types of Recommendation Systems
- Collaborative Filtering
- Content Based Filtering
- Matrix Factorization
- Pros and Cons of Collaborative Filtering
- Content Based filtering
- Hybrid Recommender System

Module-8: Time-series Forecasting

- Introduction to forecasting data
- Properties of Time Series data
- Features of Time Series data
- Markov Processes - Overview and Terminologies
- Naive, Average and Moving Average Forecasting
- Exponential Smoothing
- ARIMA Approach